Anil C. Seth

Curriculum Vitae & Publication List

Updated July 9, 2026

Office Address
Dept. of Physics & Astronomy
University of Utah
201 James Fletcher Building
115 South, 1400 East
Salt Lake City, UT 84112-0830

Contact Information
Phone: (801) 585-7793
Fax: (801) 581-4801
E-mail: aseth@astro.utah.edu
Web: http://www.physics.utah.edu/∼aseth

Education

University of Washington — June 2006
Ph.D. in Astronomy
Thesis Title: "The Formation and Evolution of Late-Type Galaxies"
Thesis Advisors: Paul Hodge & Julianne Dalcanton

University of Washington — Dec. 2001
M.S. in Astronomy

Wesleyan University — May 1998
B.A. in Physics, Astronomy & Music, with honors in Astronomy

Professional Experience

Professor, Physics & Astronomy, University of Utah — 2022 – Present

Associate Professor, Physics & Astronomy, University of Utah — 2017 – 2022

Assistant Professor, Physics & Astronomy, University of Utah — 2011 – 2017

Leave Replacement Professor, Wellesley College — 2010 – 2011
Part-time position to design and teach two courses.

CfA & OIR Postdoctoral Fellowships, Harvard-Smithsonian Center for Astrophysics — 2006 – 2011
Worked independently on topics including nuclear star clusters and resolved stellar populations.

Group

Postdocs Advised: Zixian (Purmortal) Wang (2023–2025), Karina Voggel (2016–2019), Mark den Brok (2012–2015)

Graduate Students Advised: Isabella Carlton (2025–), David Ohlson (2022–), Kameron Goold (2022–), Christian Hannah (2020–), Callie Clontz (Utah/MPIA, 2022–2025), Ben Gibson (2020–2025; co-advisor), Aishwarya Ashok (2018–2022), Antoine Dumont (2017–2022), Renuka Pechetti (2015–2020), Chris Ahn (2014–2018), Dieu Nguyen (2012–2018), Cliff Johnson (U. Washington, 2010–2015, primary advisor for thesis project)

Undergraduate and Post-Bac Students Advised: Evan Kerr (2025–), Matthew Whitaker (2023–), Zack Freeman (2022–2026), Sylvia Mesiceck (2025), Jack Nguyen (2024), Ethan Lame (2021), Aasutosh Acharya (2021), David Ohlson (2020–2021), Addie Wilkerson (ACCESS, 2020), Estephani TorresVillanueva (2019–2021), Tobin Wainer (2019–2021), Olivia Cooper (REU, 2018), Lexie Wilson (Honors Thesis, 2017), Candace Bryn (ACCESS, 2017), Timothy Bell (REU,2016–2017), Julie Tang (ACCESS, 2016), Ellie Penner (ACCESS, 2015–2016), Jordan Bulkley (2014–2017), Gina Eastes (ACCESS, 2015), Alex Deich (2014–2015), Dylan Gregersen (2013–2015), Chris DiLullo (REU, 2014), Brian Kimmig (2012–2014), Matt Wallace (2013–2014), Katharine Larsen (Wellesley/Utah, 2011–2013), Nell Byler (Wellesley, 2011)

Grants & Awards

Overview: $1.2 million in new grants since 2021. Funding includes NSF PIed, collaborative proposals, and a CAREER award, as well as numerous HST and JWST grants. Recipient of the University of Utah's Mid-Career Presidential Scholar Award and Early Career Teaching Award.

James Webb Space Telescope Guest Observer Grants 2022–Present

Astrometric monitoring of the fast stars around the intermediate-mass black hole in Omega Centauri, co-I, $54189

Weighing the Intermediate Mass Black Hole in Omega Centauri, PI, $199,403, 2025-2028

Revealing Low Luminosity Active Galactic Nuclei, PI, $244,548, 2022-2025

NSF AST AAG grant 2021–2025

Counting the Unseen: Massive Black Hole Demographics from Tidal Disrution Events, PI, $299,637.

Presidential Scholar, University of Utah 2019–2021

Honors extraordinary research and academic efforts of early to mid-career faculty, $30,000.

Hubble Space Telescope Guest Observer & Archival Grants 2011–Present

Where is the central black hole in the LMC?, PI, $46,094, 2026-2029

Probing Omega Cen's Stellar Mass Black Hole Population Through Microlensing, PI, $139,950, 2023-2026

Deciphering the Formation History of Omega Cen with a Comprehensive Stellar Kinematic and Population Dataset, $213,662, 2022–2024

Solving the Mystery of Galaxy Bulges, co-I, 54 orbits, 18 orbits, $165,197, 2017–2022

Increasing Diversity in Galaxies with BH Mass Measurements, PI, 6 orbits, $46,265, 2017–2020

A Legacy Imaging Survey of M33, co-I, 108 orbits, $54,348, 2016–2020

Improving Central Black Hole Mass Measurements in Low Mass Early Type Galaxies, PI'ed by student Dieu Nguyen, 12 orbits, $57,687, 2016–2021

NSF AST AAG grant 2018–2021

Collaborative Research: Exploring the Dark Side of NGC 5128, co-PI, $191,991

National Science Foundation 2014–2019

CAREER: Understanding the Formation of Galaxy Nuclei, PI, $598,520

Early Career Teaching Award, University of Utah $2,500 Mar. 2018

Selected Telescope Time

Where is the central black hole in the LMC? 2025-2026, Cycle 33

Hubble Space Telescope, PI, 7 orbits

Weighing the Intermediate Mass Black Hole in Omega Centauri 2024-2025, Cycle 3

James Webb Space Telescope, NIRSpec Program, PI, 26.2 hours.

The Hunt for Omega Cen⋆ 2024-2025, Cycle 11

Atacama Large Millimeter Array, PI, 13.2 hours.

Revealing Low Luminosity Active Galactic Nuclei (RevealLLAGN) 2021, Cycle 1

James Webb Space Telescope, NIRSpec & MIRI, Medium Program, PI, 31.9 hours.

The Formation History of Omega Centauri 2020–

Hubble Space Telescope, 18 orbits using WFC3 imaging, PI, 2021, Cycle 29

Very Large Telescope, MUSE instrument, 24.3 hours, co-I, 2020

Telescope time listed above is allocated through competitve proposals with oversubscription rates for PI'ed proposals ranging from 3–9. Approximate Monetary Cost of Telescope Time Awarded as PI Since Arrival in Utah – $4,880,000

Teaching Experience

Keck Lecturer in Dark Sky Studies 2019 – 2021
Helped develop interdisciplinary dark skies minor at U. of Utah. Ongoing support of minor.

Early Career Teaching Award, University of Utah Mar. 2018

Teaching as a Faculty Member, University of Utah 2012 – Present
Taught 90-person introductory astronomy course (1060; Spring 2012), 25 person observational astronomy course (2060/4060; Fall 2012), 10-14 person graduate level extragalactic astronomy course (5570; Spring 2013, 2015, 2017, Fall 2020, Spring 2024), 14-24 person observational astronomy course (2060/4060; Fall 2013, Fall 2014, Fall 2015, Fall 2018, Fall 2022), started new undergraduate extragalactic astrophysics course (Fall 2016, 2017, 2021), graduate research methods in astronomy (6410; Spring 2018, 2020,2021, 2022), instructor for new Being Human in STEM course (SCI3900; Spring 2020), Undergraduate Seminar focused on career preparation (3980; Fall 2023–Present; taught each semester), Lightscapes course on dark sky studies (3850; Fall 2023, Fall 2024), Foundations of Astrophysics (3070; Fall 2025, 34 students)

Visiting Professor, Wellesley College 2010 – 2011
Taught undergraduate seminar on stellar populations and general education astrobiology course.

U. of Utah Service

Director of Undegraduate Studies 2020–Present
Led major reform of undergruate physics program that has increased our number of majors.

Executive/Leadership Committee, Physics & Astronomy Department 2021–Present
Member of three person committee advising chair on key decisions.

Post Baccalaureate Program Steering Committee 2022–2024
Helped finalize, judge applicants, and support incoming students.

Curriculum Committees 2020–Present
Chaired departmental curriculum committee (2020-2021), part of college curriculum committee.

Department Action and Leadership Institute 2021
Led proposal to attend 2021 intensive workshop on departmental change.

Graduate Advisory Committee 2018
Helped reform the department's graduate program.

Additional Committees & Service: Student Awards Committee (2023–2026), Astronomy Faculty Search Committee (2015–2016 [chair], 2020–2021), Career Line Faculty Search Committee (2021–2022), Teaching Excellence Committee (2022–2023), South Physics Observatory Committee (2012–2019, chair), High Energy Astrophysics Seminar Committee (2012-2013, chair 2013-14, chair 2016-17), Policy Board (2014-2015), Undergraduate Advising Committee (2012 – Present), Conference for Undergraduate Women in Physics (2014), College of Science Academic Appeals Committee (2011 – 2014), College of Science Council (2017-2019), Futures Committee (2017-2018), Outreach Committee (2017-2018)

Recent Professional Service

AURA Representative 2024–
Led Utah's joining of AURA, an oversight board for astronomy facilities.
I attend yearly two-day meetings in Tucson, AZ.

Scientific Organizing Committee 2026
ESO conference on Galaxy Nuclei, Santiago, Chile.

JWST Cycle 3 Time Allocation Committee Panelist 2024

NSF Astronomy & Astrophysics Grant Panelist 2020

Sloan Digital Sky Survey V Advisory Committee 2019–Present

Referee & TAC External Reviewer 2009 – Present
Refereed papers for Nature, Nature Astronomy, AAS Journals, Monthly Notices of the RAS, James Webb Space Telescope, NSF ASCEND

Recent Talks

Recent Invited seminars or colloquiua: Minnesota (Mar. 2026), Texas A&M (Jan. 2026), Minnesota (Feb. 2025), Toronto (Mar. 2025), UC-Boulder (Oct. 2024), Utah (Oct. 2024), Weber St. (Feb. 2023), Pabna University (Jan. 2022, virtual), Edinburgh (Dec. 2020, virtual), McGill (Oct. 2019), Columbia (Oct. 2019), Washington (May 2019), MPE-Garching (Jul. 2018), Distinguished Colloquium Washington State (Mar. 2018)

Invited Review, *Galactic Nuclei Astrophysical Phenomena*, Sexten, Utaly, Jul. 2026
Invited Participant & Speaker, *KITP Program: Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys*, Santa Barbara, Dec. 2025
Invited Participant, *Galactic Centers as tracers of Galaxy Evolution*, Lorentz Center, Leiden, the Netherlands, Oct. 2025
Contributed Talk, *The Origin and Evolution of SMBHs*, Sexten, Italy Jul. 2024
Invited Talk, *European Astronomical Society Meeting*, Krakow, Poland, Jul. 2023
Invited Talk & Panelist, *Leveraging Ground- and Space-Based Facilities to Tackle Astronomy's Hardest Questions, 241st AAS*, Seattle, Washington, Jan. 2023
Invited Talk, *Origin, growth and feedback of black holes in dwarf galaxies*, San Sebastian, Spain, Sep. 2022
Contributed Talk, *Intermediate Mass Black Holes: New Science from Stellar Evolution to Cosmology*, San Juan PR, Apr. 2022
Contributed Talk, *Dynamical Formation of Gravitational Wave Sources*, Aspen, Jan. 2022
Contributed Talk, *American Astronomical Society Virtual Meeting*, Jan. 2021
Contributed Talk, *Supermassive Black Holes (organized by U. de Chile)*, Dec. 2020
Invited Review *Globular Clusters at the Nexus of Star and Galaxy Formation*, KITP (Virtual), May 2020
Contributed Talk *MODEST-20: Dense Star Clusters in the Era of Large Surveys*, Mumbai, India, Feb. 2020
Invited Review *IAU Symposium 351: Star Clusters: from the Milky Way to the Early Universe*, Bologna, Italy, May 2019
Invited Speaker *Formation of Stars and Massive clusters in Dwarf Galaxies over Cosmic Time*, Lorentz Center, Leiden, the Netherlands, Feb. 2019

Recent Outreach

Astronomy Outreach Internships at Bryce Canyon: creation, yearly astro training for interns and other rangers, fund raising, and administration, ∼6,000 directinteractions per year, 2015–Present.
Night Sky Internship started in collaboration with Capitol Reef National Park and Torrey, Utah non-profit Entrada Institute; administration, fund raising, 2023–Present
Dark Sky Talks, Torrey Town Theater, Astronomy on Tap SLC, Apr. 2026
Chair, Dark Sky Utah Board, Jan. 2026–
Invited Speaker, Capitol Reef National Park Heritage Starfest, ∼30 people, Sep. 2024
Invited Speaker, U. Utah National Advisory Council (large donor group), ∼25 people, Sep. 2024
Two (sold-out) talks at Clark Planetarium IMAX theater, ∼450 people, Aug. 2024
Press-release about Omega Centauri Black Hole, Jul. 2024, coverage in many local media outlets
U-Science summer camp, P&A volunteer – gave talk, organized demos, 25 students, Jun. 2024
Invited Talk/Panel, Utah Symposium in Science and Literature, Apr. 2024
Organized Physics Teacher Workshop, Mar. 2024
Public Talk, Utah Valley Astronomical Society & Utah Astronomy Club, Mar./Sep. 2024
Public Talk, SPS Regional Meeting, Mar. 2024
Bryce Canyon Astronomy Festival, Telescope/Daytime Volunteer, Jun. 2022–2026
Public Talk, Bryce Canyon Astronomy Festival, ∼200 people Jun. 2023

Refereed Publications

Summary: 46 group or first authored publications plus 151 co-authored publications.
Total citation count: 13903, h-index: 67, Click here for ADS listing

First-Author or Group Publications

46) RevealLAGN 1: JWST Emission-line Spectra Reveal Low-luminosity AGN with Ultraviolet-deficient Spectral Energy Distributions and Warm Molecular Gas
Goold, K., **Seth, A.**, [22 authors], 2026, ApJ, 1000, 271.

45) oMEGACat. VIII. A Subpopulation Census of ω Centauri
Clontz, C., **Seth, A.**, [9 authors], 2026, ApJ, 998, 177.

44) oMEGACat. VII. Tracing Interstellar and Intracluster Medium of ω Centauri Using Sodium Absorptions
Wang, Z., **Seth, A.**, [15 authors], 2025, ApJ, 994, 143.

43) Counting the Unseen. II. Tidal Disruption Event Rates in Nearby Galaxies with REPTiDE
Hannah, C., Stone, N., **Seth, A.**, van Velzen, S., 2025, ApJ, 988, 29.

42) oMEGACat. V. Helium Enrichment in ω Centauri as a Function of Metallicity
Clotntz, C., **Seth, A.**, [14 authors], 2025, ApJ, 984, 162.

41) oMEGACat. IV. Constraining the Ages of Omega Centauri Subgiant Branch Stars with HST and MUSE
Clotntz, C., **Seth, A.**, [14 authors], 2024, ApJ, 977, 14.

40) Counting the Unseen. I. Nuclear Density Scaling Relations for Nucleated Galaxies
Hannah, C., **Seth, A.**, Stone, N., van Velzen, S., 2024, AJ, 168, 137.

39) RevealLAGN 0: First Look at JWST MIRI Data of Sombrero and NGC 1052
Goold, K., **Seth, A.**, [29 authors], 2024, ApJ, 966, 204.

38) Investigating the dark matter halo of NGC 5128 using a discrete dynamical model
Dumont, A., **Seth, A.**, [8 authors], 2024, A&A, 685, 132.

37) The 50 Mpc Galaxy Catalog (50 MGC): Consistent and Homogeneous Masses, Distances, Colors, and Morphologies
Ohlson, A., **Seth, A.**, [3 authors], 2024, AJ, 167, 31.

36) Composite Bulges. III. A Study of Nuclear Star Clusters in Nearby Spiral Galaxies
Ashok, A., **Seth, A.**, [18 authors], 2023, ApJ, 958, 100.

35) A Population of Luminous Globular Clusters and Stripped Nuclei with Elevated Mass to Light Ratios around NGC 5128
Dumont, A., **Seth, A.**, [9 authors], 2022, ApJ, 928, 147.

34) The Panchromatic Hubble Andromeda Treasury: Triangulum Extended Region (PHATTER). III.
The Mass Function of Young Stellar Clusters in M33 Wainer, T., Johnson, C., **Seth, A.**, [7 authors], 2022, ApJ, 928, 15.

33) First direct dynamical detection of a dual supermassive black hole system at sub-kiloparsec separation
Voggel, K., **Seth, A.**, [8 authors], 2022, *A&A*, 658, 152.

32) Detection of a 100,000 $M_\odot$ black hole in M31's Most Massive Globular Cluster: A Tidally Stripped Nucleus
Pechetti, R., **Seth, A.**, [7 authors], 2022, ApJ, 924, 48.

31) Spatially Resolving the Star Formation Histories of Three Nearby Nuclear Star Clusters
Hannah, C., **Seth, A.**, [5 authors], 2021, *AJ*, 162, 281.

30) Luminosity Models and Density Profiles for Nuclear Star Clusters for a Nearby Volume-limited Sample of 29 Galaxies
Pechetti, R., **Seth, A.**, [4 authors], 2020, *ApJ*, 900, 32.

29) A Gaia-based Catalog of Candidate Stripped Nuclei and Luminous Globular Clusters in the Halo of Centaurus A
Voggel, K., **Seth, A.**, [5 authors], 2020, *ApJ*, 899, 140.

28) Surprisingly Strong K-band Emission Found in Low-luminosity Active Galactic Nuclei
Dumont, A., **Seth, A.**, [4 authors], 2020, *ApJ*, 888, 19.

27) Improved Dynamical Constraints on the Masses of the Central Black Holes in Nearby Low-mass Early-type Galactic Nuclei And the First Black Hole Determination for NGC 205
Nguyen, D., **Seth, A.**, [18 authors], 2019, *ApJ*, 872, 104.

26) The Impact of Stripped Nuclei on the Super-Massive Black Hole Number Density in the Local Universe
Voggel, K., **Seth, A.**, [4 authors], 2019, *ApJ*, 971, 159.

25) Nearby Early-type Galactic Nuclei at High Resolution:
Dynamical Black Hole and Nuclear Star Cluster Mass Measurements
Nguyen, D., **Seth, A.**, [13 authors], 2018, *ApJ*, 858, 118.

24) The Black Hole in the Most Massive Ultracompact Dwarf Galaxy M59-UCD3
Ahn, C., **Seth, A.**, [20 authors], 2018, *ApJ*, 858, 102.

23) Upper Limits on the Presence of Central Massive Black Holes
in Two Ultra-compact Dwarf Galaxies in Centaurus A
Voggel, K., **Seth, A.**, [11 authors], 2018, *ApJ*, 858, 20.

22) Detection of Enhanced Central Mass-to-Light Ratios
in Low-Mass Early-Type Galaxies: Evidence for Black Holes?
Pechetti, R., **Seth, A.**, [5 authors], 2017, *ApJ*, 850, 15.

21) Panchromatic Hubble Andromeda Treasury. XVIII.
The High-Mass Truncation of the Star Cluster Mass Function
Johnson, L. C., **Seth, A.**, [8 authors], 2017, *ApJ*, 839, 78.

20) Detection of Supermassive Black Holes in Two Virgo
Ultracompact Dwarf Galaxies
Ahn, C., **Seth, A.**, [14 authors], 2017, *ApJ*, 839, 72.

19) Improved Dynamical Constraints on the Mass of the
Central Black Hole in NGC 404
Nguyen, D., **Seth, A.**, [7 authors], 2017, *ApJ*, 836, 237.

18) Panchromatic Hubble Andromeda Treasury. XVI.
Star Cluster Formation Efficiency and the Clustered Fraction of Young Stars
Johnson, L. C., **Seth, A.**, [11 authors], 2016, *ApJ*, 827, 33.

17) Panchromatic Hubble Andromeda Treasury. XII.
Mapping Stellar Metallicity Distributions in M31
Gregersen, D., **Seth, A.**, [12 authors], 2015, *AJ*, 150, 189.

16) Measuring the Mass of the Central Black Hole in the
Bulgeless Galaxy NGC 4395 from Gas Dynamical Modeling
den Brok, M., **Seth, A.**, [8 authors], 2015, *ApJ*, 809, 101.

15) PHAT Stellar Cluster Survey. II. Andromeda Project Cluster Catalog
Johnson, L. C., **Seth, A.**, [12 authors], 2015, *ApJ*, 802, 127.

14) Measuring Consistent Masses for 25 Milky Way Globular Clusters
Kimmig, B., **Seth, A.**, [3 authors], 2015, *AJ*, 149, 53.

13) A Supermassive Black Hole in an Ultracompact Dwarf Galaxy
Seth, A., [13 authors], 2014, *Nature*, 513, 98.

12) Extended Structure and Fate of the Nucleus in Henize 2-10
Nguyen, D., **Seth, A.**, [4 authors], 2014, *ApJ*, 794, 34.

11) PHAT Stellar Cluster Survey I. Year 1 Catalog and Integrated Photometry
Johnson, L. C., **Seth, A.**, [22 authors], 2012, *ApJ* 752, 95.

10) The ACS Nearby Galaxy Survey Treasury. X.
Quantifying the Star Cluster Formation Efficiency of Nearby Dwarf Galaxies
Cook, D., **Seth, A.**, [7 authors], 2012, *ApJ*, 751, 100.

9) Gas Accretion in the M32 Nucleus: Past & Present
Seth, A., 2010, *ApJ*, 725, 670.

8) The NGC 404 Nucleus: Star Cluster and Likely Intermediate Mass Black Hole
Seth, A., Cappellari, M., Neumayer, N., Caldwell, N., [7 authors], 2010, *ApJ*, 714 713.

7) The Rotating Nuclear Star Cluster in NGC 4244
Seth, A., Blum, R., [4 authors], 2008, *ApJ*, 687, 997.

6) The Coincidence of Nuclear Star Clusters and Active Galactic Nuclei
Seth, A., Agueros, M., Lee, D., Basu-Zych, A., 2008 *ApJ*, 678, 116.

5) Clues to Nuclear Star Cluster Formation from Edge-on Spirals
Seth, A., Dalcanton, J., Hodge, P., Debattista, V., 2006 *AJ*, 132, 2539.

4) A Study of Edge-On Galaxies with HST/ACS. II. Vertical Distribution
of the Resolved Stellar Population
Seth, A., Dalcanton, J., de Jong, S., 2005 *AJ*, 130, 1574.

3) A Study of Edge-On Galaxies with HST/ACS. I. Initial results
Seth, A., Dalcanton, J., de Jong, S., 2005 *AJ*, 129, 1331.

2) Star Clusters in Virgo and Fornax Dwarf Irregular Galaxies
Seth, A., Olsen, K., Miller, B., Lotz, J., Telford, R., 2004 *AJ*, 127, 798.

1) Water Masers as Tracers of Protostellar Disks and Outflows
in the Intermediate-Mass Star-forming Region NGC 2071
Seth, A., Greenhill, L., Holder, B., 2002 *ApJ*, 581, 325.

Additional co-authored papers available at this ADS listing.